Write your questions and thoughts here!

Geometric Sequence

If each successive term in a sequence has a _____ (or constant proportional change), the sequence is called geometric. The nth term of a geometric sequence is given by

$$g_n =$$

where ____ is the initial value [zero term] and ____ is the common ratio.

4. Find an equation (rule) that represents the sequence $\{2, 6, 18, 54, \dots\}$.

You can write the equation for the sequence using ANY term, not just the initial term.

$$g_n =$$

where ____ is the ____ term of the sequence.

5. What are other ways a rule can be written for the sequence $\{2, 6, 18, 54, \dots\}$

6. Division is still geometric. It is multiplying by a fraction between 0 and 1. What is a rule for the sequence $\{64, 16, 4, 1, \dots\}$ for $k = 1$?

2.1 Change in Arithmetic and Geometric Sequences

AP Precalculus

2.1 Practice

Find an equation that gives the nth term of each sequence. Use the initial value $(k = 0)$ of the sequence in your equation.

1. $\{9, 27, 81, 243, \dots\}$	2. $\{-4, -1, 2, 5, \dots\}$	3. $\{-1, -4, -16, -64, \dots\}$

Find an equation that gives the *n*th term of each sequence. Use the initial value ($k = 0$) of the sequence in your equation.

4. $\left\{10, 5, \frac{5}{2}, \frac{5}{4}, \ldots\right\}$

5. $\{-5, -9, -13, -17, \ldots\}$

6. $\left\{\frac{4}{3}, 1, \frac{3}{4}, \frac{9}{16}, \ldots\right\}$

Find an equation that gives the *n*th term of each sequence. Instead of the initial value use the *k*th term of the sequence in your equation. *k* is given for each problem.

7. $\left\{75, 15, 3, \frac{3}{5}, \ldots\right\}$ $k = 2$

8. $\{44, 36, 28, 20, \ldots\}$ $k = 4$

9. $\{-10, -100, -1{,}000, -10{,}000, \ldots\}$ $k = 3$

10. $\{11, 22, 33, 44, \ldots\}$ $k = 2$

11. $\left\{\frac{9}{4}, \frac{3}{2}, 1, \frac{2}{3}, \ldots\right\}$ $k = 3$

12. $\left\{\frac{1}{9}, \frac{1}{3}, 1, 3, \ldots\right\}$ $k = 2$

Find the *n*th term of each sequence. Write an equation for each sequence before finding the nth term.

13. $\{105, 95, 85, 75, \ldots\}$ What is the 87th term?

14. $\left\{\frac{1}{6}, 1, \frac{11}{6}, \frac{8}{3}, \ldots\right\}$ What is the 34th term?

15. $\{-86, -80, -74, -68, \ldots\}$ What is the 52nd term?

16. $\{-21, -25, -29, -33, \ldots\}$ What is the 120th term?

17. Which of the following equations could represent the sequence $\{10, 2, -6, -14, \ldots\}$?

I. $f(n) = 10 - 8(n-1)$ II. $f(n) = -8 + 10n$ III. $f(n) = -8 + 10(n-1)$

IV. $f(n) = 18 - 8n$ V. $f(n) = 10 - 8n$ VI. $f(n) = 2 - 8(n+2)$

---

18. Which of the following equations could represent the sequence $\left\{\frac{1}{2}, 2, 8, 32, \ldots\right\}$?

I. $f(n) = \frac{1}{2}\left(\frac{1}{4}\right)^{n-1}$ II. $f(n) = \frac{1}{2}(4)^{n-1}$ III. $f(n) = 8(4)^{n+3}$

IV. $f(n) = 2\left(\frac{1}{4}\right)^{n-2}$ V. $f(n) = \frac{1}{8}(4)^{n}$

---

19. The first term of an arithmetic sequence is 4, and the common difference of the sequence is 3. What is the seventh term of the sequence?

(A) 22 (B) 25 (C) 972 (D) 2,916

2.1 Change in Arithmetic and Geometric Sequences

AP Precalculus Name: _____ **CA #1**

Find an equation that gives the *n*th term of each sequence. Use the initial value $(k = 0)$ of the sequence in your equation.

1. $\{-57, -42, -27, -12, \ldots\}$

2. $\left\{\frac{1}{5}, \frac{6}{5}, \frac{36}{5}, \frac{216}{5}, \ldots\right\}$

3. $\left\{-18, -6, -2, -\frac{2}{3}, \ldots\right\}$

Find an equation that gives the *n*th term of each sequence. Instead of the initial value use the *k*th term of the sequence in your equation. *k* is given for each problem.

4. $\{192, 48, 12, 3, \ldots\}$ $k = 1$

5. $\{-63, -66, -69, -72, \ldots\}$ $k = 3$

6. $\{0.25, 2, 16, 128, \ldots\}$ $k = 4$

Find the *n*th term of each sequence. Write an equation for each sequence before finding the nth term.

7. $\{107, 97, 87, 77, \ldots\}$ What is the 65th term?

8. $\{8, 16, 24, 32, \ldots\}$ What is the 107th term?

9. Which of the following equations could represent the sequence $\{2, 6, 10, 14, \ldots\}$?

I. $f(n) = 4n - 2$ II. $f(n) = 4n + 2$ III. $f(n) = 2n + 4$

IV. $f(n) = 2 + 4(n - 1)$ V. $f(n) = 10 + 4(n - 2)$ VI. $f(n) = -6 + 4(n + 1)$

10. Which of the following equations could represent the sequence $\left\{25, 5, 1, \frac{1}{5}, \ldots\right\}$?

I. $f(n) = 125\left(\frac{1}{5}\right)^n$

II. $f(n) = \frac{1}{5}(25)^{n-1}$

III. $f(n) = 25\left(\frac{1}{5}\right)^{n-1}$

IV. $f(n) = \frac{1}{5}(5)^{n-2}$

Answers to 2.1 CA #1

1. $a_n = -72 + 15n$	2. $g_n = \frac{1}{30}(6)^n$	3. $g_n = -54\left(\frac{1}{3}\right)^n$	4. $g_n = 192\left(\frac{1}{4}\right)^{n-1}$	5. $a_n = -69 - 3(n-3)$
6. $g_n = 128(8)^{n-4}$	7. -533	8. 856	9. I, IV, and VI.	10. I and III.

AP Precalc

Write your questions and thoughts here!

2.2 Change in Linear and Exponential Functions

2.2 Notes

Linear Functions vs. Arithmetic Sequences

Slope-intercept form of a linear function	Arithmetic Sequence
$f(x) =$	$a_n = a_0 + dn$

Initial Value $=$ Constant rate of change $=$

When not starting with the initial value, we can transform the above equations by using the *k*th term. Our arithmetic sequence is…

Point-slope form of a linear function	Arithmetic Sequence with a known *k*th term
$f(x) =$	$a_n = a_k + d(n-k)$

The point (x_1, y_1) for a linear function is similar to the term (k, a_k) of an arithmetic sequence.

Exponential Functions vs. Geometric Sequences

Exponential Function	Geometric Sequence
$f(x) =$	$g_n = g_0 r^n$

When not starting with the initial value, we can transform the above equations by using the *k*th term. Our geometric sequence is

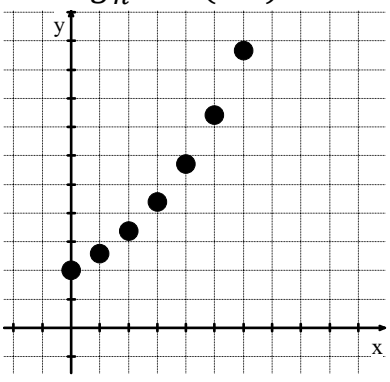
Shifted Exponential Function	Geometric Sequence with a known *k*th term
$f(x) =$	$g_n = g_k r^{(n-k)}$

The point (x_1, y_1) for an exponential function is similar to the term (k, g_k) of a geometric sequence.

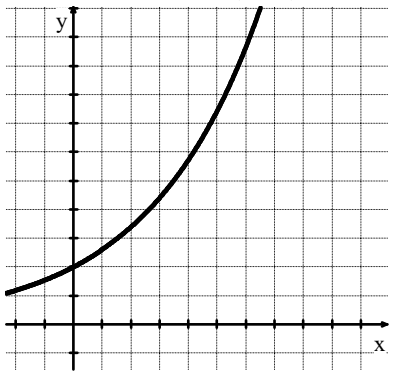
Domain

Remember the domain of a sequence won't be the same as the domain of its corresponding function.

$g_n = 2(1.3)^n$



$f(x) = 2(1.3)^x$



The domain of this sequence is _____ The domain of this function is _____

Write your questions and thoughts here!

Linear functions and exponential functions can both be expressed analytically in terms of an initial value and a constant involved with change. There is a difference between the two though. Linear functions are based on addition. Exponential functions are based on multiplication.

Recognizing how to change the output values

Over equal-length input-value intervals, if the output values of a function change…

_____, then the function is linear. (ADDING the SLOPE)

_____, then the function is exponential. (MULTIPLYING the RATIO)

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

1. $(2, 1), (3, 4), (4, 16)$

2. $(5, 3), (6, 7), (7, 10)$

3. $(3, -1), (5, 7), (7, 15)$

If we know a function is linear or exponential (for sequences that would be arithmetic or geometric), then you only need two distinct values to come up with an equation (rule) for the function or sequence.

4. It is known that $f(x)$ is a linear function and that it passes through the points $(3, 7)$ and $(8, 1)$. Write an equation for this function.

5. It is known that $f(x)$ is an exponential function and that it passes through the points $(2, 5)$ and $(4, 12)$. Write an equation for this function.

The following functions are either linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

6. $y = 4 \cdot 2^x$

7. $y = 5x + 7$

The following functions are either linear or exponential. Which is it? Justify your answer.

8.

x	3	7	11
$f(x)$	10	6	2

9.

x	1	3	5
$f(x)$	2	6	18

2.2 Change in Linear and Exponential Functions

AP Precalculus

2.2 Practice

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

1. $(1,2), (2,8), (3,20)$

2. $(10,30), (11,20), (12,10)$

3. $(13,20), (14,4), \left(15, \frac{4}{5}\right)$

4. $(7,6), (8,12), (9,20)$

The following functions are either linear or exponential. Which is it? Justify your answer.

5.

x	1	5	9
$f(x)$	2	6	18

6.

x	-4	-2	0
$f(x)$	5	1	-3

7.

x	7	10	13
$f(x)$	5	105	205

8.

x	11	20	29
$f(x)$	8	4	2

Is each function linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

9. $y = -\frac{1}{4} \cdot 7^x$

10. $y = 4x - 6$

11. $y = 2 - 9x$

12. $y = \frac{1}{3} \cdot \left(\frac{2}{5}\right)^x$

13. $y - 7 = -5(x + 4)$

14. $y + 1 = \left(\frac{3}{4}\right)^{x+7}$

15. $y + 5 = \frac{1}{6} \cdot 2^{x-3}$

16. $y - 6 = 3(x - 10)$

It is known that $f(x)$ is a linear function and that it passes through the given points. Write an equation for this function.		
17. $(1, 4)$ and $(3, 10)$	18. $(4, 15)$ and $(10, 3)$	19. $(10, 2)$ and $(13, 5)$

It is known that $f(x)$ is an exponential function and that it passes through the given points. Write an equation for this function.		
20. $(1, 4)$ and $(3, 10)$	21. $(4, 15)$ and $(10, 3)$	22. $(10, 2)$ and $(13, 5)$

2.2 Change in Linear and Exponential Functions

2.2 Test Prep

23. Mr. Brust has been collecting He-Man figures for 40 years. The number of figures he owns can be modeled by an arithmetic sequence, where the first year is year 1. The number of figures in year 5 was 12, and the number of figures in year 20 was 167. How many He-Man figures did he have in year 14?

24. **Calculator active.** Mr. Bean has lots of siblings and lots of nephews and nieces. The number of people in his family can be modeled using a geometric sequence, where the first generation is generation 1. The number of people after generation 2 is 52. The number of people after generation 4 is 468. How many people will be in the Bean family after generation 6?

25. The third term of a sequence is 5, and the fifth term of the sequence is 20. Of the following, which statement is true?

    (A) If the sequence is arithmetic, the first term could be -25.

    (B) If the sequence is arithmetic, the fourth term could be 7.5.

    (C) If the sequence is geometric, the fourth term could be 7.

    (D) If the sequence is geometric, the sixth term could be 40.

2.2 Change in Linear and Exponential Functions

AP Precalculus Name: _____ **CA #1**

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

1. $(-2, 5), (-1, 8), (0, 11)$

2. $(8, 10), (9, 12), (10, 8)$

3. $(2, 12), (3, 6), (4, 3)$

The following functions are either linear or exponential. Which is it? Justify your answer.

4.

x	-3	2	7
$f(x)$	1	3	9

5.

x	2	6	10
$f(x)$	12	9	6

Is each function linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

6. $y = -4 \cdot \left(\frac{1}{3}\right)^x$

7. $y = 6 + 2x$

8. $y = 8 \cdot 3^{x+6}$

9. $y + 1 = 6\left(x - \frac{2}{3}\right)$

It is known that $f(x)$ is a linear function and that it passes through the given points. Write an equation for this function.

10. $(8, 12)$ and $(10, 6)$

11. $(3, 5)$ and $(8, 40)$

It is known that $f(x)$ is an exponential function and that it passes through the given points. Write an equation for this function.

12. $(8, 12)$ and $(10, 6)$

13. $(3, 5)$ and $(8, 40)$

Answers to 2.2 CA #1

1. linear	2. neither	3. exponential	4. Exponential because for each input change of 5, f changes proportionally by a ratio of 3.	
5. Linear because for each input change of 4, f changes at a constant rate -3.	6. Ratio of $\frac{1}{3}$	7. Slope of 2	8. Ratio of 3	9. Slope of 6
10. $y - 12 = -3(x - 8)$ or $y - 6 = -3(x - 10)$	11. $y - 5 = 7(x - 3)$ or $y - 40 = 7(x - 8)$	12. $y = 12 \cdot \left(\sqrt{\frac{1}{2}}\right)^{x-8}$ or $y = 6 \cdot \left(\sqrt{\frac{1}{2}}\right)^{x-10}$	13. $y = 5 \cdot \left(\sqrt[5]{8}\right)^{x-3}$ or $y = 40 \cdot \left(\sqrt[5]{8}\right)^{x-8}$	

AP Precalc

Write your questions and thoughts here!

2.3 Exponential Functions

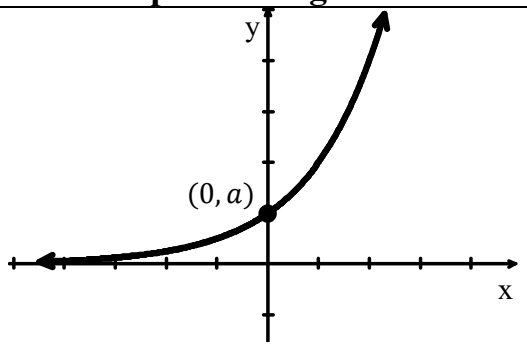
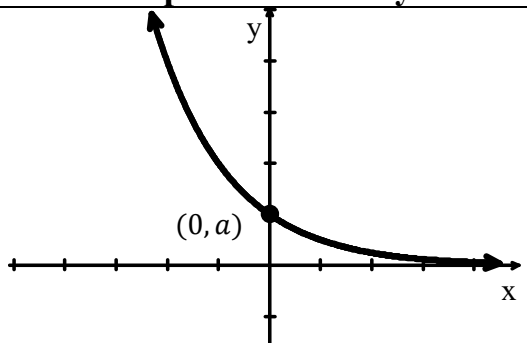
2.3 Notes

General Form of an Exponential Function

An exponential function has the general form

$$f(x) =$$

Where the initial value is ___. Also, _____ and _____.

Exponential growth	Exponential decay
	
when $a > 0$ and $b > 1$.	when $a > 0$ and $0 < b < 1$.

Identify if the exponential function is growth or decay and justify your response.

1. $f(x) = 2(0.6)^x$
   Exponential Growth or Decay

2. $f(x) = 2\left(\frac{7}{5}\right)^x$
   Exponential Growth or Decay

3. $f(x) = \frac{4}{3}\left(\frac{1}{6}\right)^x$
   Exponential Growth or Decay

4. $f(x) = 0.2(1.3)^x$
   Exponential Growth or Decay

Examine the function $f(x) = 3(2)^x$. What are we doing to the initial value 3 when the input value is $x = 5$?

$$f(5) = 3(2)^5$$
$$f(x) = 3 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$$

The following values are output values of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value.

5. $4 \cdot 4 \cdot 4 \cdot 1.5$

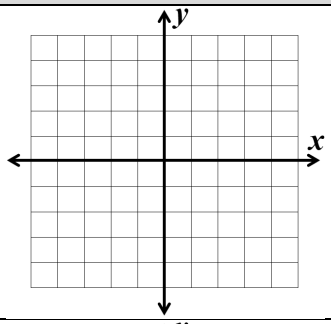
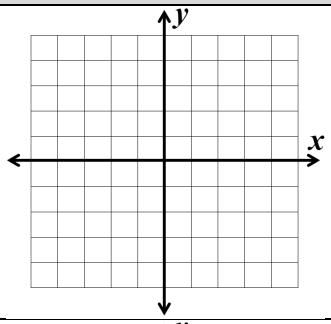
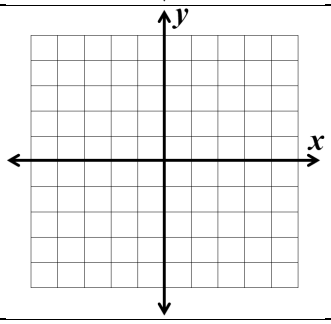
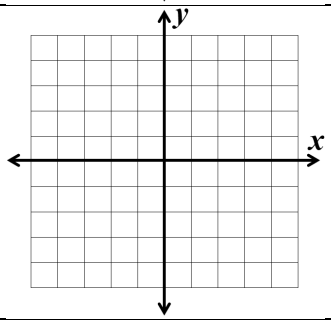
   $f(x) =$ where $x =$

6. $0.8 \cdot 3.2 \cdot 0.8$

   $f(x) =$ where $x =$

Write your questions and thoughts here!

Using your knowledge of exponential functions, sketch the general shape of each exponential function with the given parameters.

	$b > 1$	$0 < b < 1$
$a > 0$		
$a < 0$		

Characteristics of Exponential Functions

Exponential Functions have the following characteristics.

- They are **always increasing** or **always decreasing**.
  - There are no extrema. (Except on a closed interval.)
- Their graphs are **always concave up** or **always concave down**.
  - Their graphs do not have inflection points.
- If the input values increase or decrease without bound, the end-behavior can be expressed as

$$\lim_{x\to\pm\infty} ab^x = \quad \text{or} \quad \lim_{x\to\pm\infty} ab^x = \quad \text{or} \quad \lim_{x\to\pm\infty} ab^x =$$

Answer the questions for each exponential function.

7. $f(x) = 5(0.9)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

8. $f(x) = -6(2)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

Additive Transformation of an Exponential Function

Let $g(x) = f(x) + k$. This just means we take a function f and add something to it to shift the graph up or down. The function f is having k added to it to create g.

If the output values of g are proportional over equal-length input-value intervals, then f is exponential.

2.3 Exponential Functions

AP Precalculus

2.3 Practice

Identify if the function is exponential growth or decay and justify your response.

1. $f(x) = 9(3.1)^x$

Exponential Growth or Decay

2. $f(x) = 6.8(0.4)^x$

Exponential Growth or Decay

3. $f(x) = 2.1(0.06)^x$

Exponential Growth or Decay

4. $f(x) = 8\left(\frac{11}{5}\right)^x$

Exponential Growth or Decay

5. $f(x) = 1.5\left(\frac{3}{4}\right)^x$

Exponential Growth or Decay

6. $f(x) = \frac{7}{6}\left(\frac{6}{7}\right)^x$

Exponential Growth or Decay

7. $f(x) = \frac{12}{17}\left(\frac{17}{12}\right)^x$

Exponential Growth or Decay

8. $f(x) = 18(5.6)^x$

Exponential Growth or Decay

The following values are output values of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value.

9. $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 7$

$f(x) =$

where $x =$

10. $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 6$

$f(x) =$

where $x =$

11. $5 \cdot 5 \cdot 5$

$f(x) =$

where $x =$

12. $(-2) \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

$f(x) =$

where $x =$

Answer the questions for each exponential function.

13. $f(x) = 7(2)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x \to -\infty} f(x) =$

d. Find $\lim_{x \to \infty} f(x) =$

14. $f(x) = -4(5)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x \to -\infty} f(x) =$

d. Find $\lim_{x \to \infty} f(x) =$

15. $f(x) = (0.2)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x \to -\infty} f(x) =$

d. Find $\lim_{x \to \infty} f(x) =$

16. $f(x) = -6(0.8)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x \to -\infty} f(x) =$

d. Find $\lim_{x \to \infty} f(x) =$

17. $f(x) = 6\left(\frac{1}{9}\right)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

18. $f(x) = -(0.4)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

2.3 Exponential Functions

2.3 Test Prep

19.

x	0	1	2	3	4
$f(x)$	40	20	10	5	$\frac{5}{2}$

The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. The table gives values of $f(x)$ at selected values of x. Which of the following statements is true?

(A) f demonstrates exponential decay because $a > 0$ and $0 < b < 1$.

(B) f demonstrates exponential decay because $a > 0$ and $b > 1$.

(C) f demonstrates exponential growth because $a > 0$ and $0 < b < 1$.

(D) f demonstrates exponential growth because $a > 0$ and $b > 1$.

---

20. The function h is a function of the form $h(x) = a \cdot b^x$, where $a \neq 0$ and $b > 1$. The function h is also given by $h(x) = f(x) + 2$. Which of the following statements is true.

(A) The output values of both f and h are proportional over equal-length input-value intervals.

(B) The output values of f only, not h, are proportional over equal-length input-value intervals.

(C) The output values of h only, not f, are proportional over equal-length input-value intervals.

(D) The output values of neither f nor h are proportional over equal-length input-value intervals.

2.3 Exponential Functions

AP Precalculus Name: _____ **CA #1**

Identify if the function is exponential growth or decay and justify your response.

1. $f(x) = 2.5\left(\frac{2}{3}\right)^x$	2. $f(x) = \frac{5}{3}\left(\frac{3}{5}\right)^x$	3. $f(x) = 5.7(0.2)^x$	4. $f(x) = 8(2.1)^x$
Exponential Growth or Decay	Exponential Growth or Decay	Exponential Growth or Decay	Exponential Growth or Decay

The following values are output values of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value.

5. $7 \cdot 7 \cdot 7 \cdot 3.4$	6. $0.8 \cdot 0.8 \cdot 5.2$	7. $6 \cdot 1.3 \cdot 1.3 \cdot 1.3 \cdot 1.3$	8. $9 \cdot 9 \cdot 9$
$f(x) =$	$f(x) =$	$f(x) =$	$f(x) =$
where $x =$	where $x =$	where $x =$	where $x =$

Answer the questions for each exponential function.

9. $f(x) = \left(\frac{1}{5}\right)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

10. $f(x) = 3(4.5)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

11. $f(x) = -5(0.8)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

12. $f(x) = -2(6)^x$

a. Is the function increasing or decreasing?

b. Is the function concave up or concave down?

c. Find $\lim_{x\to-\infty} f(x) =$

d. Find $\lim_{x\to\infty} f(x) =$

Answers to 2.3 CA #1

1. Decay because $a > 0$ and $0 < b < 1$	2. Decay because $a > 0$ and $0 < b < 1$	3. Decay because $a > 0$ and $0 < b < 1$	4. Growth because $a > 0$ and $b > 1$
5. $f(x) = 3.4(7)^x$ where $x = 3$	6. $f(x) = 5.2(0.8)^x$ where $x = 2$	7. $f(x) = 6(1.3)^x$ where $x = 4$	
8. $f(x) = 9^x$ where $x = 3$	9. a. Decreasing b. Concave up c. $\lim_{x\to-\infty} f(x) = \infty$ d. $\lim_{x\to\infty} f(x) = 0$	10. a. Increasing b. Concave up c. $\lim_{x\to-\infty} f(x) = 0$ d. $\lim_{x\to\infty} f(x) = \infty$	11. a. Increasing b. Concave down c. $\lim_{x\to-\infty} f(x) = -\infty$ d. $\lim_{x\to\infty} f(x) = 0$	12. a. Decreasing b. Concave down c. $\lim_{x\to-\infty} f(x) = 0$ d. $\lim_{x\to\infty} f(x) = -\infty$

AP Precalc

Write your questions and thoughts here!

2.4 Exponential Function Manipulation

2.4 Notes

Recall: The **Negative Exponent Property** states that $b^{-n} =$

Recall: The **Product Property** for exponents: $b^m b^n =$

$$2^2 \cdot 2^3 = 2$$

Let's think of a **horizontal translation** of an exponential function. Remember, horizontal means we are doing something the x-values. $f(x) = b^{(x+k)}$, where b is a constant.

This now becomes a vertical dilation!

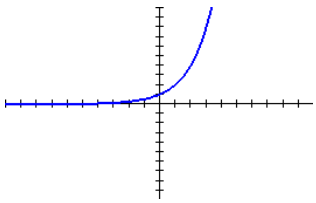
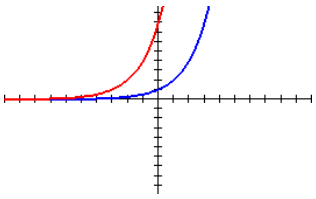
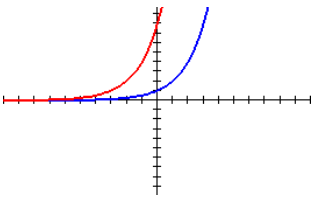
Horizontal Translation

Every horizontal translation of an exponential function, $f(x) = b^{(x+k)}$, is equivalent to a

_____.

$$f(x) = b^x \cdot b^k = b^x b^k = ab^x \text{ where } a = b^k$$

Compare the graphs.

Original function	Horizontal translation	Vertical dilation
$f(x) = 2^x$	$g(x) = 2^{(x+3)}$	$g(x) = 8(2)^x$
		

Let $f(x) = 3^x$. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

1. $g(x) = 9 \cdot 3^x$

2. $g(x) = \frac{3^x}{27}$

3. $g(x) = \frac{f(x)}{9}$

Recall: The **Power Property** for exponents: $(b^m)^n =$

$$(2^2)^3 = 2$$

Let's think of a **horizontal dilation** on an exponential function. Remember, horizontal means we are doing something the x-values. $f(x) = b^{(cx)}$, where b and c are constants.

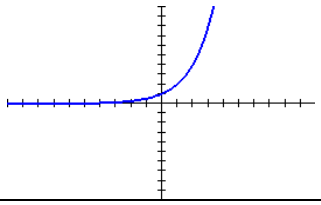
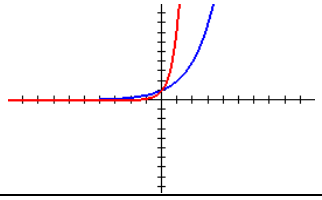
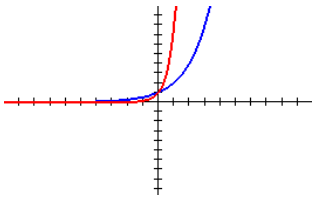
Write your questions and thoughts here!

Horizontal Dilation

Every horizontal dilation of an exponential function, $f(x) = b^{(cx)}$, is equivalent to a _____ _____ of an exponential function.

$f(x) = (b^c)^x$ where b^c is a constant and $c \neq 0$.

Compare the graphs.

Original function	Horizontal dilation	Change of base
$f(x) = 2^x$	$g(x) = 2^{(3x)}$	$g(x) = 8^x$
		

Let $f(x) = 3^x$. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

4. $g(x) = 9^x$

5. $g(x) = \big(f(x)\big)^{-3}$

The kth root of b

$b^{(1/k)}$ is the kth root of b.

$$b^{(1/k)} =$$

where k is a natural number and the root exists.

Evaluate the function at the given input values.

6. Let $h(x) = 2 \cdot 5^{x/2}$. Find $h(1)$

7. Let $h(x) = 6 \cdot 2^{x/3}$. Find $h(-2)$

2.4 Exponential Function Manipulation

AP Precalculus

2.4 Practice

Let $f(x)$ be a function on which a transformation occurs. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

1. $f(x) = 3^x$ and $g(x) = f(x) \cdot 27$

2. $f(x) = 9^x$ and $g(x) = 3^x$

3. $f(x) = 7^x$ and $g(x) = \frac{1}{7^x}$

4. $f(x) = 4^x$ and $g(x) = f(x) \cdot 16$

5. $f(x) = 4^x$ and $g(x) = \frac{f(x)}{16}$.

6. $f(x) = 5^x$ and $g(x) = \big(f(x)\big)^{-3}$

7. $f(x) = 6^x$ and $g(x) = \frac{f(x)}{36}$

8. $f(x) = 3^x$ and $g(x) = 27^x$

9. $f(x) = 4^x$ and $g(x) = -(16)^x$

10. $f(x) = 16^x$ and $g(x) = 2^x$

11. $f(x) = 7^x$ and $g(x) = -\frac{f(x)}{49}$

12. $f(x) = 2^x$ and $g(x) = \frac{f(x)}{16}$

13. $f(x) = 3^x$ and $g(x) = -3f(x)$

14. $f(x) = 2^x$ and $g(x) = f(x) \cdot 32$

Evaluate the function at the given input values.

15. Let $h(x) = 2 \cdot 3^{x/2}$. Find $h(1)$

16. Let $h(x) = 4 \cdot 4^{x/5}$. Find $h(2)$

17. Let $h(x) = 7 \cdot 2^{x/4}$. Find $h(-2)$

18. Let $h(x) = 2 \cdot 6^{x/3}$. Find $h(-1)$

2.4 Exponential Function Manipulation

2.4 Test Prep

19. **Calculator active.** A lake in the Cascade Mountains has frozen over during the winter. As spring brings warmer weather, the ice sheet begins to melt. The table below gives the area of the ice, is square feet, at various times, in days since the beginning of spring.

Time (days)	0	1	2
Area (millions of square feet)	2.5	2	1.6

The area of the ice can be modeled by the function $A(t) = ab^t$, where $A(t)$ is the area, in square feet, at time t days since the beginning of spring.

a. Use the given data to write two equations that can be used to find the values for constants a and b in the expression $A(t)$.

b. Find the values for a and b, then write the expression for $A(t)$.

c. Use the given data to find the average rate of change of the area from $t = 0$ to $t = 2$ days. Show the computations that lead to your answer.

d. Interpret the meaning of your answer from part c in the context of the problem.

e. Use the average rate of change found in part c to estimate the area of the ice, in square feet, at time $t = 3$ days. Show the work that leads to your answer.

f. Does the model found in Part b demonstrate exponential growth or exponential decay? Give a reason for your answer.

2.4 Exponential Function Manipulation

AP Precalculus Name: _____ **CA #1**

Let $f(x)$ be a function on which a transformation occurs. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

1. $f(x) = 2^x$ and $g(x) = \frac{f(x)}{32}$

2. $f(x) = 5^x$ and $g(x) = f(x) \cdot 125$

3. $f(x) = 11^x$ and $g(x) = \big(f(x)\big)^{-1}$

4. $f(x) = 7^x$ and $g(x) = 49^x$

5. $f(x) = 6^x$ and $g(x) = \big(f(x)\big)^{-4}$

6. $f(x) = 6^x$ and $g(x) = f(x) \cdot 36$

7. $f(x) = 16^x$ and $g(x) = 2^x$

8. $f(x) = 2^x$ and $g(x) = -2f(x)$

9. $f(x) = 3^x$ and $g(x) = \frac{f(x)}{27}$

Evaluate the function at the given input values.

10. Let $h(x) = 5 \cdot 2^{x/2}$. Find $h(1)$

11. Let $h(x) = 2 \cdot 3^{x/5}$. Find $h(2)$

12. Let $h(x) = 5 \cdot 4^{x/4}$. Find $h(-2)$

13. Let $h(x) = 6 \cdot 7^{x/3}$. Find $h(-1)$

Answers to 2.4 CA #1

1. Shift right 5 units.	2. Shift left 3 units.	3. Reflection across the y-axis.	4. Horizontal dilation
5. Horizontal dilation and reflection across the y-axis.	6. Shift left 2 units.	7. Horizontal dilation	8. Reflection across the x-axis and vertical dilation by a factor of 2.
9. Shift right 3 units.	10. $5\sqrt{2}$	11. $2\sqrt[5]{9}$	12. $\frac{5}{\sqrt[4]{16}} = \frac{5}{2}$	13. $\frac{6}{\sqrt[3]{7}}$

AP Precalc

Write your questions and thoughts here!

2.5.A Exponential Function Context and Data Modeling

2.5.A Notes

Recall: Exponential functions model growth patterns with successive output values over equal-length input-value intervals are proportional.

Sometimes, looking at a data set you will not see a proportional growth pattern even though it is an exponential function. Look at the following table of values.

x	0	1	2	3	4
$f(x)$	-2	1	7	19	43

It's hard to see the growth pattern for these output values. But if we add 5 to the output values, it becomes more obvious.

x	0	1	2	3	4
$f(x)$	3	6	12	24	48

1. Below is a table of values for an exponential function in the form $f(x) = a(b)^x + k$. Write the equation for this exponential function.

x	0	1	2	3	4
$f(x)$	6	12	36	132	516

The number e

$$e \approx 2.718$$

The natural base e is often used as the base in exponential functions that model contextual scenarios.

Write your questions and thoughts here!

Exponential Regression

2. The table gives the amount Mr. Kelly owes Mr. Sullivan each year after he loses in fantasy football. No payments have been made, so Mr. Kelly's debt grows each year with interest being applied.

Year (t)	1	2	3	4	5
Debt (y) (dollars)	12	20	59	109	267

a. Use an exponential regression $y = ab^x$ to model these data. Round to three decimals but store the original equation in your calculator.

b. Use the model (stored in your calculator) to predict how much Mr. Kelly will owe Mr. Sullivan in Year 7, assuming he continues to lose.

c. If Mr. Kelly loses every year from now on, and refuses to pay Mr. Sullivan, in what year will Mr. Sullivan be owed $5,000?

2.5.A Exponential Function Context and Data Modeling

AP Precalculus

2.5.A Practice

Below is a table of values for exponential functions in the form $f(x) = a(b)^x + k$. Write the equation that represents each table.

1.

x	0	1	2	3	4
$f(x)$	2	10	34	106	322

2.

x	0	1	2	3	4
$f(x)$	8	13	23	43	83

3.

x	0	1	2	3	4
$f(x)$	5	29	149	749	3,749

---

4. The table presents values of the function f for selected values of x.

x	-8	-7	-1	2
$f(x)$	30	40	158	280

a. Find an exponential regression $y = ab^x$ to model these data. Round to three decimals but store the original equation in your calculator.

b. Use the model stored in your calculator to predict the value of $f(0.5)$.

c. According to the model, when will $f(x) = 100$?

d. According to the model, when will $f(x) = 1{,}000$?

5. The table gives the number of cells c of a culture of bacteria found in a petri dish after t days.

Days (t)	3	5	10	13
Cells (c)	300	350	966	1,807

a. Use an exponential regression $c(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.

b. According to the model, how many cells were in the culture of bacteria after 8 days? Use the model stored in your calculator, not the rounded answer from part a.

c. Use the model stored in your calculator to predict the number of bacteria cells after 31 days.

d. How many days will it take for the culture to have 1 million cells?

---

6. The table gives the population of a city, in millions, for selected years where $t = 2$ represents 2012 and $t = 5$ represents 2015.

Year	2012	2015	2018	2021
Population (millions)	0.801	0.933	1.36	1.85

a. Use an exponential regression $f(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.

b. Use the model stored in your calculator to predict the population of the city in 2030.

c. During what year did the population reach 1.5 million?

d. If the model holds, during what year will the population reach 3 million?

7. Mr. Bean had a caffeinated soda at 8:00 a.m. The amount of caffeine (measured in mg) in Mr. Bean's system t hours is shown in the table below.

Hours (t)	1	3	8	12	13
Caffeine (mg)	128	85	47	20	18

a. Use an exponential regression $C(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.

b. Use the model stored in your calculator to predict the amount of caffeine still in Mr. Bean's body after 15 hours.

c. How long will it take before the caffeine level is 5 mg?

d. How long will it take before the caffeine level is 1 mg?

2.5.A Exponential Function Context and Data Modeling

2.5.A Test Prep

8. **Calculator active.** In the xy-plane, the graphs of the linear function L and the exponential function E both pass through the points $(0, 4)$ and $(1, 8)$. The exponential function E is in the form $E(x) = ab^x$. The function f is given by $f(x) = E(x) - L(x)$. What is the minimum value of f.

2.5.A Exponential Function Context and Data Modeling

AP Precalculus Name: _____ **CA #1**

Below is a table of values for exponential functions in the form $f(x) = a(b)^x + k$. Write the equation that represents each table.

1.

x	0	1	2	3	4
$f(x)$	7	16	34	70	142

2.

x	0	1	2	3	4
$f(x)$	4	13	49	193	769

3. Mr. Sullivan left out his Pico-Chile-Mucho-Caliente salsa, and it is starting to grow mold. The table below gives the area of the mold, in square centimeters, at time t days.

Day (t)	1	3	6	8
Area (square centimeters)	0.5	2	10	23

a. Use an exponential regression $A(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.

b. According to the model in your calculator, how large will the surface of the mold be after 7 days?

c. According to the model in your calculator, how large will the surface of the mold be after 10 days?

d. When will the area of the mold reach 34 cm^2?

4. Mr. Kelly has agreed to take over The Algebros social media accounts, and he feels the best way to grow it is by regularly posting his infinite amount of dad jokes. Below is a table of values for the number of followers The Algebros have for selected values of m, in months, since Mr. Kelly takes over the accounts.

Month	1	4	9	14	24
Followers	500,000	310,000	200,000	150,000	75,000

a. Use an exponential regression $F(m) = ab^m$ to model these data. Round to three decimals but store the original equation in your calculator.

b. According to the model in your calculator, how many followers will The Algebros have after 1 year? (round to the nearest whole number)

c. When will there be 100,000 followers?

d. When will there be 50,000 followers?

Answers to 2.5.A CA #1

1. $f(x) = 9(2)^x - 2$	2. $f(x) = 3(4)^x + 1$	3a. $y = 0.335(1.725)^x$ 3b. 15.233 3c. 78.164 4d. $x = 8.473$ days	4a. $y = 459{,}604.662(0.925)^x$ 4b. 179,498 4c. 19.466 months 4d. 28.313 months

AP Precalc

Write your questions and thoughts here!

2.5.B Exponential Function Context and Data Modeling

2.5.B Notes

Percent Increase:

Start with the initial amount "a" and grow with a % increase.

$$y = a(\qquad\qquad)^x$$

Percent Decrease:

Start with the initial amount "a" and decay with a % decrease.

$$y = a(\qquad\qquad)^x$$

Identify the percent increase or decrease of each function.

1. $f(x) = 0.5(1.2)^x$
2. $f(x) = 2(0.7)^x$
3. $f(x) = 4(0.025)^x$

For each problem, create a function to model the scenario.

4. A home's value V is \$200,000 and increases by 1.5% per year t.
5. A population p of fruit flies is currently at 150 flies but is increasing by 205% every day d.
6. Mr. Brust's yard is getting overrun with weeds. The first year he bought his home, there was 1800 square feet of grass g. It is decreasing by 28.1% per year t.

.

7. The value V of a new motorcycle purchased for \$11,000 decreases by 15.7% per year t.

Half-Life:

An initial amount $\boldsymbol{a}$ that shrinks by half every $\boldsymbol{h}$.

$$\boldsymbol{f(t) =}$$

Doubling time:

An initial amount $\boldsymbol{a}$ that doubles every $\boldsymbol{d}$.

$$\boldsymbol{f(t) =}$$

Write your questions and thoughts here!

For each problem, create a function to model the scenario.	
8. There is 150 grams g of radioactive material. Its half-life is 4,400 years t.	9. Mr. Bean's parents currently have 95 descendants. The number of their descendants D doubles every 15 years t.

Equivalent forms

Equivalent forms of an exponential model can reveal different properties. For example…

If t represents number of weeks, then the base of $f(t) = (1.01)^t$ indicates that the quantity increases by a factor of 1.01 every week.

The equivalent form $f(t) = (1.01^{52})^{(t/52)}$ indicates that the quantity increases by a factor of 1.01^{52} **every year**.

10. If $f(d) = 3^d$ indicates that the quantity increases by a factor of 3 every day, then what does $f(t) = (3^7)^{(d/7)}$ indicate?

2.5.B Exponential Function Context and Data Modeling

AP Precalculus

2.5.B Practice

Identify the percent increase or decrease of each function.		
1. $f(n) = 0.2(1.9)^n$	2. $f(x) = 19(0.4)^x$	3. $f(n) = 5.2(3.302)^n$
4. $f(x) = 801(0.965)^x$	5. $f(x) = 58(0.8)^x$	6. $f(n) = 10(2.061)^n$

For each problem, create a function to model the scenario.		
7. Mr. Sullivan has 24 basketballs in his garage. The number of basketballs b increases at a rate of 37.6% per year t.	8. There are 1,000 fish f in a pond that are dying off at a rate of 2.5% per month m.	9. Plutoniu-238 has a half-life of 88 years t. A scientist is studying a 3-gram sample s.

10. A stock portfolio has a value v of \$2,400 and doubles every 13 years t.

11. Mr. Brust bought a new trailer. It cost him \$35,000. Unfortunately, its value v depreciates in value by 6.5% per year t.

12. A new disease is killing the local frog population. The half-life of the population p is 2 weeks w. There are currently 6,000 frogs in the swamp.

13. Mr. Kelly's house has 8 mice running around. Their population p increases at a 450% rate per month m.

14. The number of people p living in Sullyana doubles every 130 years t. There is currently one person living there.

15. A population p of fruit flies increases by 14.34% each day d. There were 20 fruit flies in the house when you left.

16. Mr. Kelly's youngest child breaks the dishes on a regular basis. If Mr. Kelly never buys new dishes, then the number of dishes d he owns has a half-life of 10 months m. He currently has 60 dishes.

17. 900 grams of radioactive material m decays at a rate of 2.6% per year t.

18. A culture of bacteria has 500 cells c that doubles every 4 hours h.

For each of the problems below, identify how the equivalent form reveals a different property.

19. If $f(m) = 5^m$ indicates that the quantity increases by a factor of 5 every month, then what does $f(m) = (5^{12})^{(m/12)}$ indicate?

20. If $f(d) = 1.2^d$ indicates that the quantity increases by a factor of 1.2 every day, then what does $f(d) = (1.2^7)^{(d/7)}$ indicate?

21. If $f(s) = 1.02^s$ indicates that the quantity increases by a factor of 1.02 every second, then what does $f(s) = (1.02^{60})^{(s/60)}$ indicate?

22. If $f(t) = 2^t$ indicates that the quantity increases by a factor of 2 every year, then what does $f(t) = (2^{100})^{(t/100)}$ indicate?

23. On June 1, a fast-growing species of algae is accidentally introduced into a lake in a city park. It starts to grow and cover the surface of the lake in such a way that the area it covers doubles every day. If it continues to grow unabated, the lake will be totally covered and the fish in the lake will suffocate. At the rate it is growing, this will happen on June 30.

    a. Write an explicit formula for the sequence that models the percentage, p, of the surface area of the lake that is covered in algae t days after the algae was introduced. Let the initial percentage be P_0.

    b. When will the lake be covered halfway?

    c. On June 26, a pedestrian who walks by the lake every day warns that the lake will be completely covered soon. Her friend just laughs. Why might her friend be skeptical of the warning?

    d. On June 29, a cleanup crew arrives at the lake and removes almost all of the algae. When they are done, only 1% of the surface is covered with algae. How well does this solve the problem of the algae in the lake?

---

24. Zebra mussels are an invasive, fingernail-sized mollusk that are infesting some of the fresh water lakes in North America. Their population increases on average at a rate of 5% per day. Let 700 be the amount of zebra mussels in a certain lake at time $d = 0$ days. Which of the following functions g models the amount of zebra mussels after t weeks where 700 is the amount of zebra mussels at time $t = 0$?

    (A) $g(t) = 700(1.05)^{(t/7)}$

    (B) $g(t) = 700(1.05)^{7t}$

    (C) $g(t) = 700\left(1.05^{(1/7)}\right)^{(7t)}$

    (D) $g(t) = 700\left(1.05^{(7)}\right)^{(t/7)}$

---

25. Inflation is affecting the cost of goods and services in an economy at a rate of 0.667% per month. Let the cost of a particular good be worth \$100 at time $t = 0$ years. Which of the following functions gives the value of this good t years later?

    (A) $a(t) = 100(1.667^{12})^t$

    (B) $b(t) = 100\left(1 + 0.667^{(12)}\right)^t$

    (C) $c(t) = 100\left(1.00667^{(12)}\right)^t$

    (D) $d(t) = 100\left(1 + 0.00667^{(12)}\right)^t$

2.5.B Exponential Function Context and Data Modeling

AP Precalculus Name: _____ **CA #1**

Identify the percent increase or decrease of each function.

1. $f(x) = 10(0.985)^x$	2. $f(n) = 1.4(2.85)^n$	3. $y = 6(1.516)^x$	4. $f(x) = 10(0.2)^x$

For each problem, create a function to model the scenario.

5. A house sold for \$350,000. Housing prices p are expected to increase 2.1% per year t.	6. The number of visitors v to flippedmath.com doubles every 7 months m. There are currently 120,000 visitors to the website.	7. Mr. Kelly's IQ (I) is currently 105, but it is decaying at a rate of 5.8% every decade d.
8. The rodent population p in a large city is being controlled by a new poison that kills half the population every 2 months m. There are currently 1,000,000 rodents in the city.	9. A virus is spreading through the United States. On day zero, there are 121 cases c who have the virus, but it spreads at a rate of 582% increase every day d.	10. There are 3 cockroaches c behind Mr. Brust's microwave, and their population doubles every 14 days d.

11. The value v of a new motorcycle purchased for \$11,000 decreases by 15.7% per year t.	12. There is 207 grams g of radioactive material. Its half-life is 6,000 years t.

For each of the problems below, identify how the equivalent form reveals a different property.

13. If $f(m) = 1.1^m$ indicates that the quantity increases by a factor of 1.1 every minute, then what does $f(m) = (1.1^{60})^{(m/60)}$ indicate?	14. If $f(t) = 3^t$ indicates that the quantity increases by a factor of 3 every month, then what does $f(t) = (3^{12})^{(t/12)}$ indicate?

Answers to 2.5.B CA #1

1. 1.5% decrease	2. 185% increase	3. 51.6% increase	4. 80% decrease
5. $p(t) = 350{,}000(1.021)^t$	6. $v(m) = 120{,}000(2)^{m/7}$	7. $I(d) = 105(0.942)^d$	
8. $p(m) = 1{,}000{,}000\left(\frac{1}{2}\right)^{m/2}$	9. $c(d) = 121(6.82)^d$	10. $c(d) = 3(2)^{d/14}$	
11. $v(t) = 11{,}000(0.843)^t$		12. $g(t) = 207\left(\frac{1}{2}\right)^{t/6000}$	
13. The quantity increases by a factor of 1.1^{60} every hour.		14. The quantity increases by a factor of 3^{12} every year.	

AP Precalc

Write your questions and thoughts here!

2.6 Competing Function Model Validation

2.6 Notes

A quadratic and exponential curve can look similar. How do you tell if a quadratic model or an exponential model would be a better fit for a set of data? For those students familiar with statistics, you know we could look at the value of r and r^2. We are NOT doing that in this course. We are keeping it much simpler.

One way of knowing what model is best, is to examine **contextual clues**. Is something growing at a constant rate (linear)? Is it growing at a linear rate of change (rate is changing at a constant rate = quadratic)? Is it growing at a proportional rate (exponential)?

1. A pizza is removed from the oven and cools down for 5 minutes. During this time interval, the temperature is recorded once every 10 seconds. The recorded temperature with respect to time can be modeled by linear, quadratic, or exponential functions. For each of the three models, they appear to represent the data. Which of the following conclusions is best?

   (A) A linear model is best based on contextual clues.

   (B) A quadratic model is best based on contextual clues.

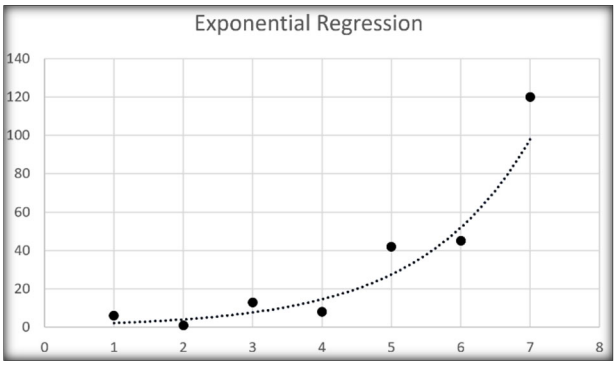
   (C) An exponential model is best based on contextual clues.

   (D) Contextual clues fail to help in selecting a model for this contextual situation.

Another very useful way to know if a model is appropriate for the data set is to look at the plots of **residuals**. But first….

What is a residual plot?

Recall that regression lines are lines/curves that "fit" a set of data.

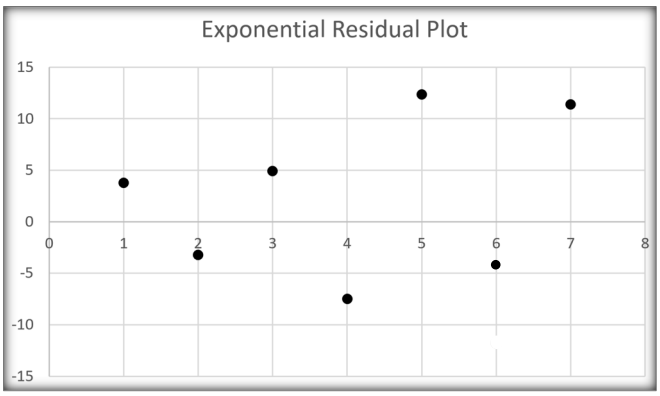


A **residual value** is a measure of how much a regression curve vertically misses a data point. You take the "actual" measured data point and subtract the "predicted" value from the regression line.

ACTUAL – PREDICTED =

Write your questions and thoughts here!

This gives us a new set of data that can be graphed as a residual plot.



The difference between the predicted and actual values is the _____ in the model. Depending on the data set and context, it may be more appropriate to have an underestimate or overestimate for any given interval.

PREDICTED – ACTUAL =

The line of best fit will give either an overestimate or an underestimate.

***Notice** that a "residual" and an "error" are very similar, and often used synonymously. There is a difference though. The sign of the error tells us if the prediction (from the regression model) is an overestimate or an underestimate.

2. A high school student is getting ready for basketball season by getting extra practice before school. Each morning, he records the number of practice shots he took, and records how many he made. The table below represents his recordings.

Shots taken t (input)	100	75	120	200	50	300
Shots made m (output)	35	22	37	60	20	126

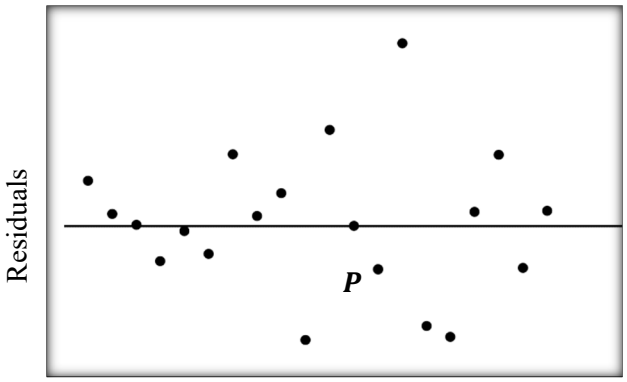
a. Find a linear regression curve.

b. Use your model from part a to find the error at $t = 50$.

c. Is the value predicted an overestimate or underestimate of the actual value?

Write your questions and thoughts here!

3. Several cars were tested for their braking distance. A regression model was calculated with the car's weight (in pounds) as input values and the braking distance (in feet) as the output values. The given residual plot has a point labeled P at the coordinate $(3{,}300, -15)$. What does point P indicate in the context?

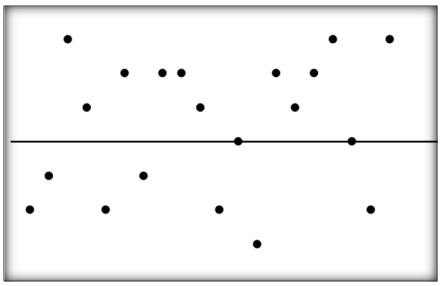


Residual Plot Patterns

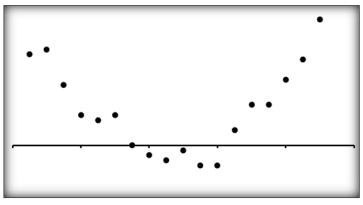
How does the shape of a residual plot tell us if the regression curve is an appropriate model for the data set? One way is to ask the question,

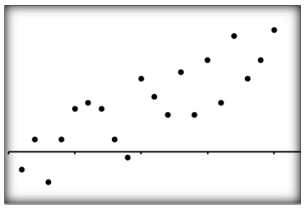
Do the residuals exhibit a clear pattern?

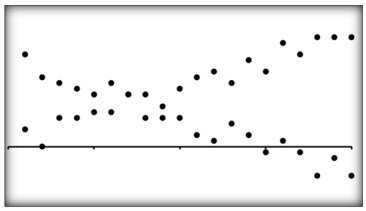
1. In a "good" residual plot, the residuals exhibit no clear pattern.
   No pattern = good regression model.



2. In a "bad" residual plot, the residuals exhibit some type of pattern such as a curve or a wave.
   Pattern = bad regression model.







2.6 Competing Function Model Validation

2.6 Practice

AP Precalculus

1. The salary of a high school teacher typically increases as they have more experience. Below is a list of teacher salaries for one district based on their years of experience.

Years taught t	0	4	10	15	25	30
Annual Salary S	\$53,300	\$61,500	\$68,800	\$72,150	\$77,600	\$79,900

   a. Find a linear regression curve. Round to three decimal places.

   b. Use your model from part a to find the error at $t = 4$.

   c. Is the value predicted an overestimate or underestimate of the actual value?

2. The altitude of the Sun is simply a measurement of how high it appears above the horizon, in terms of an angular measurement. On Halloween, Mr. Bean does his best to measure the sun's altitude throughout the day. The measurements in hours h since sunrise and the angle a in degrees are given below.

Hours h since sunrise (input)	0.5	3	7	10	11
Altitude a in degrees (output)	6	32	41	12	0

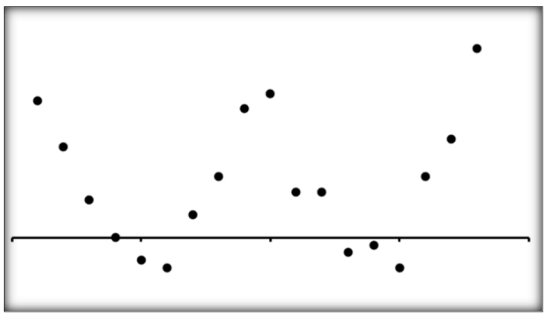
   a. Find a quadratic regression curve. Round to three decimal places.

   b. Use your model from part a to find the error at $h = 10$.

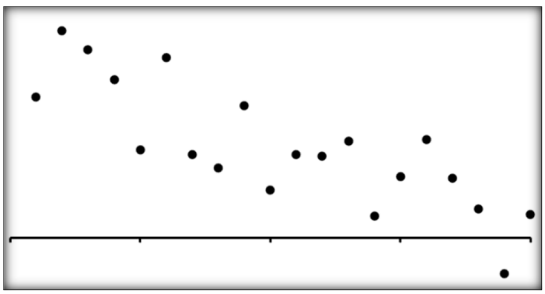
   c. Is the value predicted an overestimate or underestimate of the actual value?

3. A student creates a model for a data set. Which of the following could be an appropriate residual plot for their **linear** regression? More than one answer may apply.

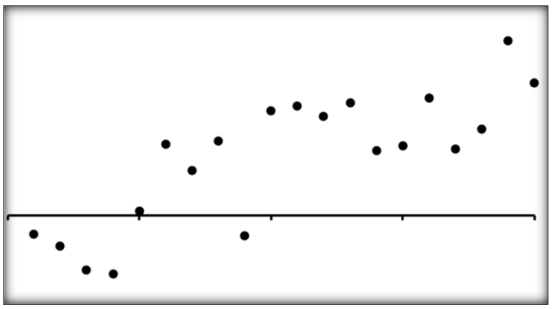
I.



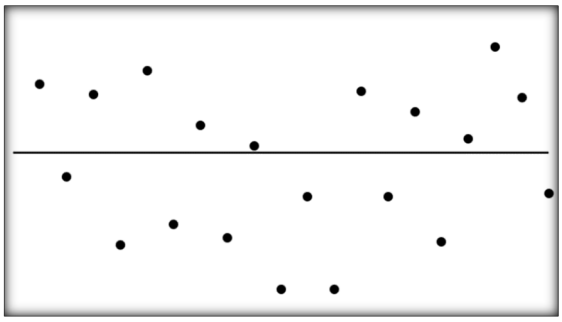
II.



III.

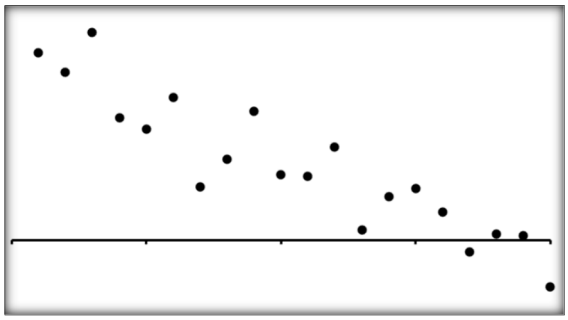


IV.

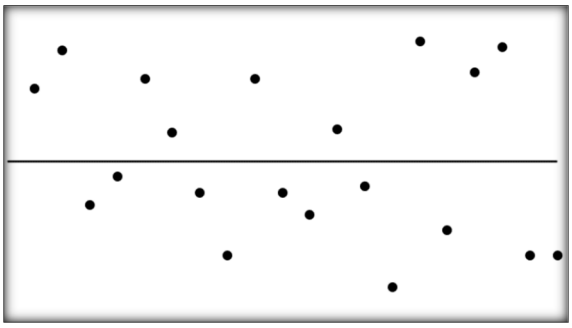


4. A student creates a model for a data set. Which of the following could be an appropriate residual plot for their **quadratic** regression? More than one answer may apply.

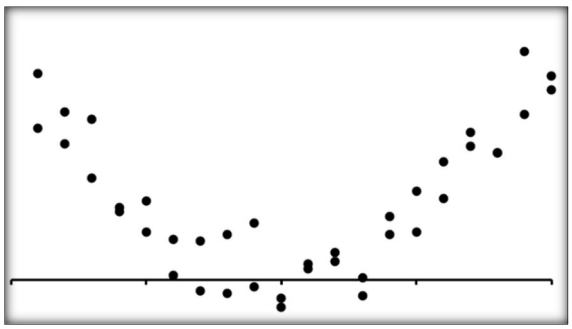
I.



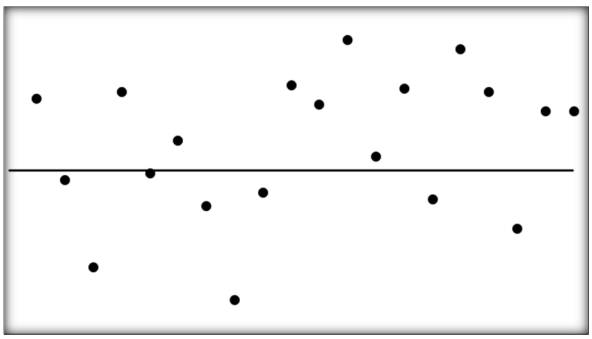
II.



III.

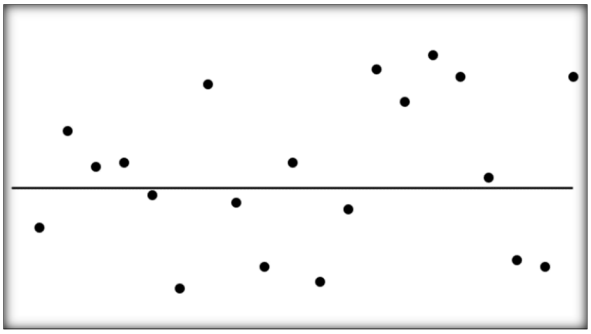


IV.

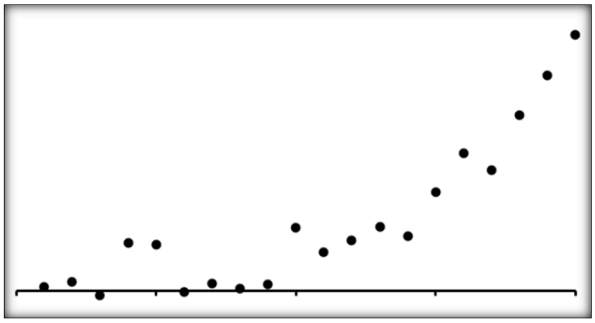


5. A student creates a model for a data set. Which of the following could be an appropriate residual plot for their **exponential** regression? More than one answer may apply.

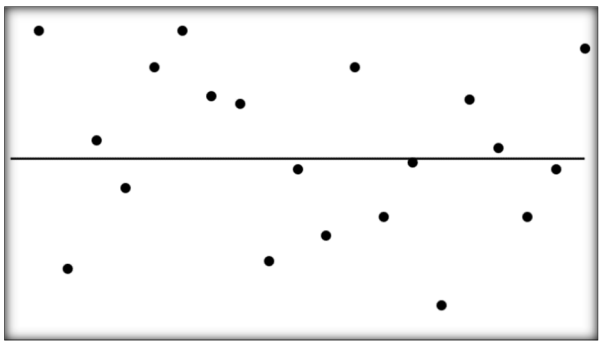
I.



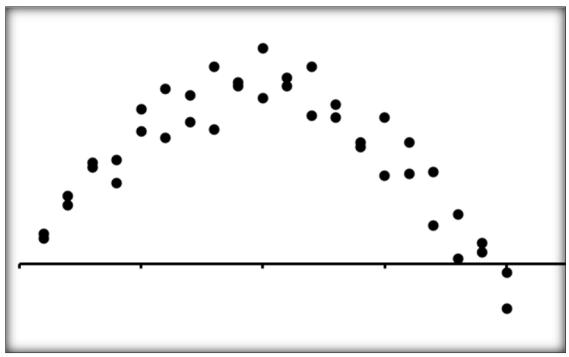
II.



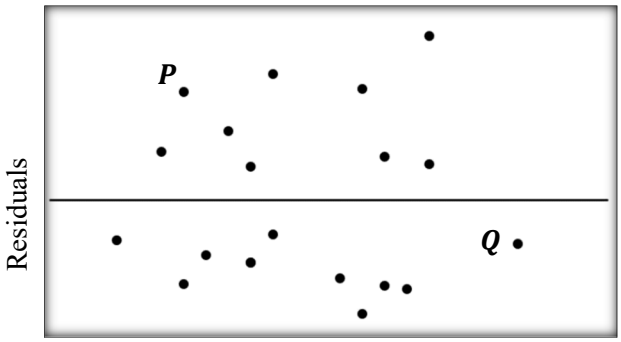
III.



IV.



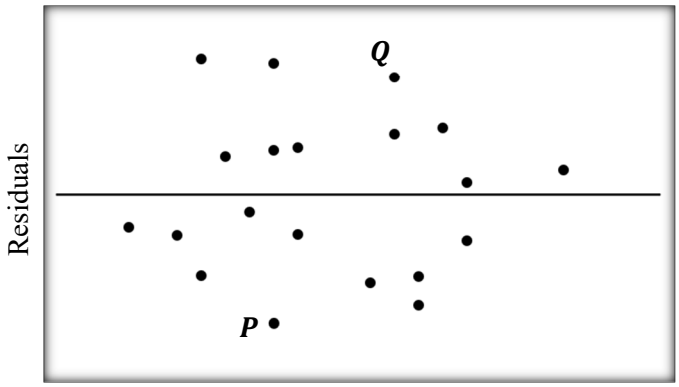
6. Students from an AP Precalculus class were brought to the gym to shoot free throws, but only the students with little-to-no basketball experience were allowed to shoot. Each student was given 60 seconds to shoot as many free throws as possible. A regression model was calculated with the student's height (in inches) as input values and the number of shots made as the output values. The residual plot for the regression model and data points is below.



a. The given residual plot has a point labeled P at the coordinate $(62, 4.7)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(74, -2.4)$. What does point Q indicate in the context?

7. After getting a driver's license, the reality of monthly insurance costs (premiums) sets in for many people. Insurance companies charge less for insurance the more years of experience a driver has. A local high school statistics class collects information from several insurance companies regarding their monthly insurance premiums. A regression model was calculated with the driver's experience (in years) as input values and the monthly premiums (in dollars) as the output values. The residual plot for the regression model and data points is below.



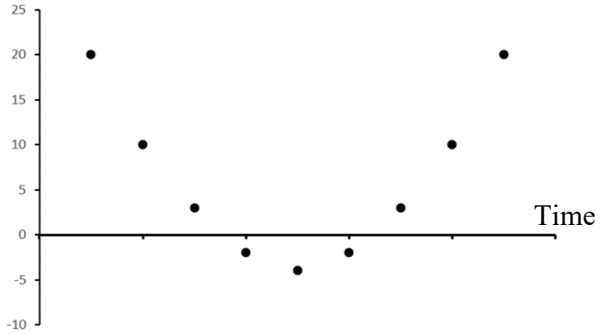
a. The given residual plot has a point labeled P at the coordinate $(6, -102.5)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(10, 97.34)$. What does point Q indicate in the context?

2.6 Competing Function Model Validation

2.6 Test Prep

8. Mr. Brust started a new online business selling Brust-merch. He made estimates about his revenue and used an exponential regression to develop a model for sales over time. The figure shows the graph of the residuals of the exponential regression. Which of the following statements about the exponential regression is true?



(A) The exponential model is appropriate because there is a clear pattern in the graph of the residuals.

(B) The exponential model is appropriate because there are data points both above and below the x-axis.

(C) The exponential model is not appropriate, because there is a clear pattern in the graph of the residuals.

(D) The exponential model is not appropriate because the residuals are not in an exponential pattern.

2.6 Competing Function Model Validation

AP Precalculus Name: _____ **CA #1**

1. The owners of a local taco stand are trying to increase revenue and decide to invest a portion of their profits to advertising. The table below shows the amount spent on advertising compared to the revenue brought in for various months of the year.

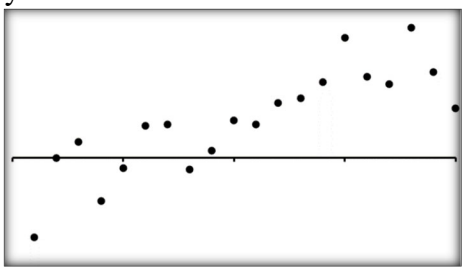
Money spent on advertising a	\$500	\$300	\$700	\$500	\$0	\$200
Revenue R	\$7,000	\$4,000	\$7,500	\$6,000	\$3,000	\$4,500

   a. Find a linear regression curve. Round to three decimal places.

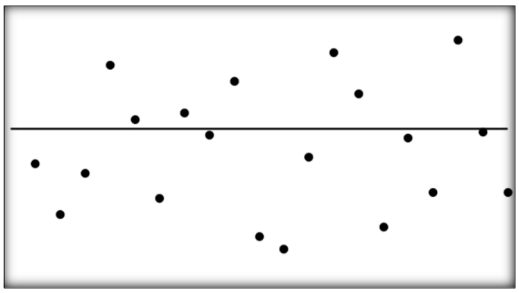
   b. Use your model from part a to find the error at $a = 200$.

   c. Is the value predicted an overestimate or underestimate of the actual value?

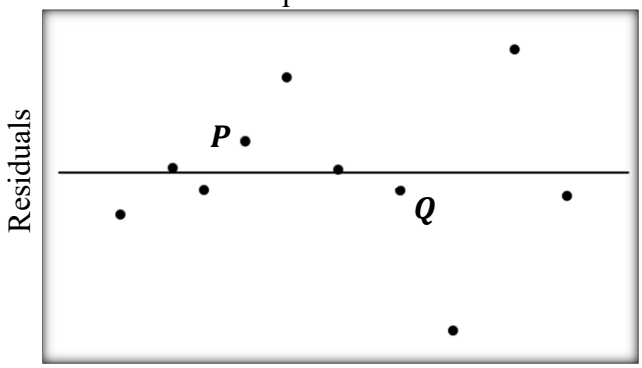
2. A student creates a model for a data set. The residual plot for their linear regression is shown. Is the model appropriate? Why or why not?



3. A student creates a model for a data set. The residual plot for their exponential regression is shown. Is the model appropriate? Why or why not?



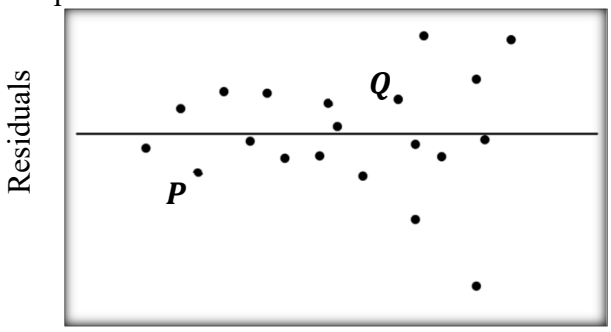
4. A doctor's office has collected data regarding patients' age and their cholesterol level. A regression model was calculated with the age as input values and the cholesterol level as the output values. The residual plot for the regression model and data points is below.



a. The given residual plot has a point labeled P at the coordinate $(43, 8.73)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(58, -5.02)$. What does point Q indicate in the context?

5. Teen birthrates (the number of births per 1000 females aged 15 to 17 years-old) are being compared to poverty levels (which is the percent of the state's population living in households with incomes below the federally defined poverty level). A regression model was calculated for several states with teen birthrate as input values and the poverty level (as a percent) as the output values. The residual plot for the regression model and data points is below.



a. The given residual plot has a point labeled P at the coordinate $(14, -2.08)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(37, 1.87)$. What does point Q indicate in the context?

Answers to 2.6 CA #1

1a. $R(a) = 6.755a + 2856.383$	1b. $4207.383 - 4500 = -292.617$	1c. Underestimate
2. No, the residual plot creates a pattern	3. Yes. No clear pattern.	
4a. Because point P is above the x-axis, for the patient with an age of 43, the model underestimates the actual cholesterol level by 8.73.	4b. Because point P is below the x-axis, for the patient with an age of 58, the model overestimates the actual cholesterol level by 5.02.	
5a. Because point P is below the x-axis, for the state with a teen birthrate of 14 per 1000 females aged 15 to 17 years-old, the model overestimates the poverty level by 2.08 percent.	5b. Because point Q is above the x-axis, for the state with a teen birthrate of 37 per 1000 females aged 15 to 17 years-old, the model underestimates the poverty level by 1.87 percent.	

AP Precalc

Write your questions and thoughts here!

2.7A Composition of Functions (Part 1)

2.7A Notes

The **composition of functions** means we take one function and substitute it into the other function. $f(g(x))$ is read as f of g of x. It means the output of $g(x)$ is used as the input for $f(x)$.

1. Let $f(x) = x^2 - 4$ and $g(x) = 3 - x$.
   a. Find $f(g(5))$.
   b. Find $g(f(5))$.

$f(g(x))$ can also be written as _____

Use the table to find the values of the composed functions below.

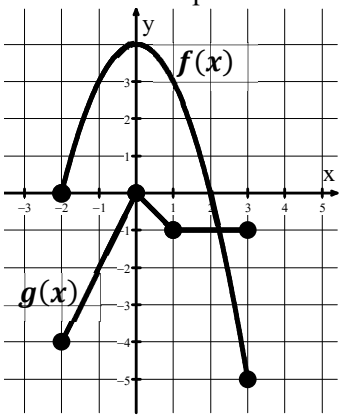
x	$f(x)$	$g(x)$
-5	3	-2
-3	0	7
-2	3	-5
0	2	3
2	-3	0
3	2	6

3. $f(g(0))$

4. $f \circ g(-2)$

5. $g \circ f(2)$

Use the graphs of f and g to find the values of the composed functions.



6. $f(g(1))$

7. $g(f(-2))$

8. $g \circ f(1)$

Write your questions and thoughts here!

Recall: The additive identity is 0. Any number + 0 gives you the same number.

Recall: The multiplicative identity is 1. Any number * 1 gives you the same number.

Identity Function

The identify function is $f(x) = x$. $f(x) = x$ composed with any function g is the same function g.

$$\text{If } f(x) = x, \text{ then } f\big(g(x)\big) =$$

2.7A Composition of Functions (Part 1)

AP Precalculus

2.7A Practice

Let $f(x) = 2 - x$ and $g(x) = \sqrt{3 - x}$.

1. Find $(f \circ g)(-2)$.
2. Find $f\big(g(2)\big)$.
3. Find $g\big(f(8)\big)$.

Let $f(x) = 4^x$ and $g(x) = x - x^2$.

4. Find $f\big(g(-1)\big)$.
5. Find $g\left(f\left(\frac{1}{2}\right)\right)$.
6. Find $(g \circ f)(-1)$.

Let $f(x) = \frac{1}{x^2}$ and $g(x) = 2^x$.

7. Find $(f \circ g)(-1)$.
8. Find $g\left(f\left(\frac{1}{2}\right)\right)$.
9. Find $g\big(f(2)\big)$.

10. Let $f(x) = 1 - 2x$ and $h(x) = f\big(g(x)\big)$. Fill in the table.

x	$g(x)$	$h(x)$
-3	-2	
-1	1	
0	2	

11. Let $g(x) = 3^x$ and $h(x) = g\big(f(x)\big)$. Fill in the table.

x	$f(x)$	$h(x)$
1	-2	
3	0	
5	1	

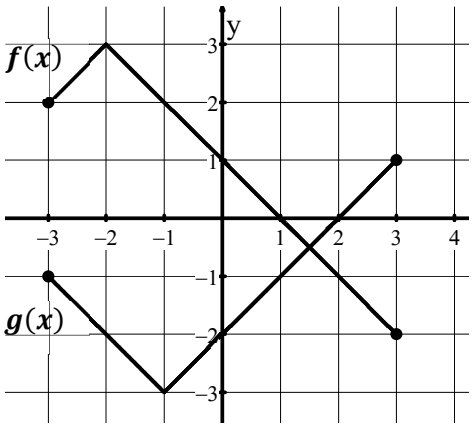
12. Let $f(x) = x^2 - 4x$ and $h(x) = f\big(g(x)\big)$. Fill in the table.

x	$g(x)$	$h(x)$
-5	1	
-3	2	
-1	-2	

13. Fill in the following table, given that $h(x) = f\big(g(x)\big)$.

x	$g(x)$	$f(x)$	$h(x)$
-3	0	7	
-2	-2	-4	
-1	-3	-1	
0	1	3	
1	2	0	
2	3	4	
3	-1	9	

14. Use the graph to find the each value.

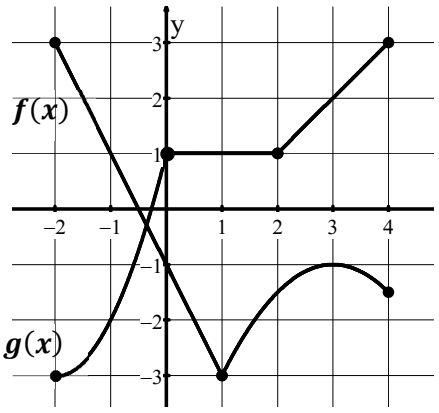


a. Find $f\big(g(-2)\big)$.

b. Find $f \circ g(0)$.

c. Find $g\big(f(3)\big)$.

15. Use the graph to find the each value.



a. Find $f\big(g(0)\big)$.

b. Find $g\big(f(3)\big)$.

c. Find $g \circ f(-2)$.